Stroma World Spec — §4: World State Variables

> **What this section defines:** Every variable the simulation tracks, at every scale. **Completeness rule:** If a variable isn't listed here, it doesn't exist in v1.0. If a module reads or writes something, that something is in this list. **Organization:** Three tiers — cell-level (per agent), local environment (per voxel), host-level (global scalars). Each variable earns its place by being necessary for at least one Reality Anchor.

Tier 1: Cell-Level Variables

Tracked per individual cell agent. Every tumor cell and every stromal cell carries these. Updated each time step according to module logic.

1.1 Identity & Lineage

Variable	Type	Tumor	Stromal	Mutable?	Description & Justification
cell_id	Integer	✓	✓	No	Unique identifier. Required for tracking individual cell fate through the simulation — specifically for Anchor 10 (spatial sanctuary), where we need to ask "which specific cells survived treatment and where were they?" Without per-cell ID, survivor analysis collapses to population averages that can't distinguish spatial clustering from uniform attrition.
cell_type	Enum {TUMOR, PSC, CAF}	✓	✓	Once	Actor class. PSC → CAF is a one-way transition (see Actors doc). TUMOR never changes type. This variable gates which behavioral rules apply:

Variable	Type	Tumor	Stromal	Mutable?	Description & Justification
					tumor cells run Modules 1–2, 4–5, 7; stromal cells run Modules 3, 6; both are affected by Module 8. Without it, the simulation can't enforce the fundamental asymmetry that tumor cells secrete and stromal cells build.
position	(x, y) or (x, y, z)	✓	✓	Yes	Spatial coordinates. Updated by motility (Module 5 for tumor EMT cells) and by PhysiCell's mechanics solver (pushing, adhesion). Position is the variable that connects every cell to its local microenvironment — the cell samples all field values at its position. Every Reality Anchor with a spatial prediction (5, 6, 9, 10) is meaningless without tracked positions.
parent_id	Integer	✓	✓	No	ID of the cell that divided to create this one. Set at birth, never changes. Required for lineage tracing during post-hoc analysis: did the sanctuary survivors in Anchor 10 descend from a common ancestor, or are they independently trapped? Not read by any module during the run — purely for analysis.

1.2 Genotype (Fixed)

Variable	Type	Tumor	Stromal	Mutable?	Description & Justification
gene_state[21]	Array of	✓	Partial	No	The 21-gene panel from the Gene Set doc. Each gene is in exactly one of three states, fixed at cell

Variable	Type	Tumor	Stromal	Mutable?	Description & Justification
	{WT, GOF, LOF}				creation, inherited by daughter cells. Stromal cells carry a subset (ACTA2, GLI1, HAS2, COL1A1, TGFB1, SHH — all WT by default; SHH = LOF because stroma doesn't secrete it). This array is the root input to the Boolean network. Without it, there is no genotype-to-phenotype mapping — the combination rules (Gene State Representation doc) have nothing to combine. Every module ultimately traces back to a gene state read.

Why fixed? Gene states represent structural mutations, not expression dynamics. KRAS is constitutively active because the protein is permanently locked in its GTP-bound form, not because expression fluctuates. Expression-level dynamics are handled by the Boolean activity variables below. Stochastic mutation during the run (clonal evolution) is Cycle 3.

1.3 Boolean Activity States (Dynamic)

These are the "is this gene doing something right now?" variables, as distinct from "is this gene intact?" (§1.2). Updated every time step by module logic.

Variable	Type	Tumor	Stromal	Module	Description & Justification
hif1a_active	Boolean	✓	✓	M1	Hypoxia-inducible factor 1α activation state. ON when local O_2 is below the threshold set by Knob 6a. This single Boolean is the bridge between the physical microenvironment (O_2 field) and the cell's adaptive response. It feeds into secretion amplification (E02), EMT induction (E01), death resistance (E04), proliferation modulation (E03), and stress priming (E05). Without it, hypoxia is just a number in a field — it never changes cell behavior. Required for Anchors 5, 6, 10.
zeb1_active	Boolean	✓	—	M5	EMT master switch. ON when induction signal (TGF-β SMAD-independent arm + HIF1A contribution) exceeds threshold (Knob 5a). Determines whether the cell is epithelial or mesenchymal. Controls motility speed, adhesion strength, MMP2 activity, and the go-vs-grow proliferation penalty. Without this variable, there is no EMT — and without EMT, there is no spatially restricted invasive front (Anchor 6), no MMP2-driven local ECM degradation, and no PD-L1 upregulation at the tumor-stroma interface.
cdh1_expressed	Boolean	✓	—	M5	E-cadherin expression. Inverse of ZEB1: ON when ZEB1=OFF, OFF when ZEB1=ON. Could be derived on-the-fly from ZEB1, but tracking it explicitly makes adhesion calculations cleaner — PhysiCell's cell_cell_adhesion_strength reads

Variable	Type	Tumor	Stromal	Module	Description & Justification
					CDH1 state directly. Required for the motility/adhesion axis of the EMT phenotype.
acta2_active	Boolean	—	✓	M3	α-SMA activation. The PSC → CAF switch. OFF = quiescent stellate cell (no ECM production, no TGF-β feedback). ON = activated fibroblast (full barrier-building behavior). Irreversible in v1.0. This is the most consequential binary state change in the simulation — it's the moment the stroma starts forming. Required for Anchor 1 (self-assembly). Every downstream stromal behavior (ECM production, TGF-β feedback, CAF proliferation) is gated by this variable.
gli1_active	Boolean	—	✓	M3	Hedgehog pathway effector in stromal cells. ON when local SHH exceeds activation threshold. Unlike ACTA2, this is **reversible** — GLI1 drops if SHH disappears (e.g., under SHH inhibition). Boosts ECM production and CAF proliferation (E11, E13). Required for Anchor 3 (SHH paradox): when SHH is blocked, GLI1 turns OFF in all CAFs, reducing but not eliminating barrier production (because the TGF-β/ACTA2 arm persists independently). Without tracking GLI1 separately from ACTA2, you can't model partial stroma depletion.
nrf2_active	Boolean	✓	—	M7	Oxidative/xenobiotic stress response. ON when the cell experiences drug-induced or oxidative stress above a threshold. Priming step before

Variable	Type	Tumor	Stromal	Module	Description & Justification
					ABCB1 activation. Required because ABCB1 doesn't activate instantly — NRF2 must turn on first, then ABCB1 follows after the induction delay (Knob 7a). Without NRF2 as a separate state, the temporal dynamics of resistance induction collapse: efflux would either be instant (too pessimistic) or impossible (too optimistic).
abcb1_active	Boolean	✓	—	M7	Drug efflux pump. ON after NRF2 activation + induction delay elapsed. When ON, reduces intracellular drug concentration by factor set by Knob 7b. This is the variable that distinguishes drug-naive cells from drug-adapted cells. Required for Anchor 10: sanctuary survivors who were never exposed to drug have ABCB1=OFF and remain drug-naive (still vulnerable if drug eventually reaches them), while cells that survived partial drug exposure have ABCB1=ON and are hardened. This distinction only exists if ABCB1 is tracked per-cell.

Variable	Type	Tumor	Stromal	Module	Description & Justification
mmp2_active	Boolean	✓	—	M5	Matrix metalloproteinase 2 activity. ON when EMT extent ≥ MEDIUM. Causes local ECM degradation at the cell's position (E19). Required for Anchor 6 (invasive front remodels local ECM) and Anchor 7 (tumor-intrinsic ECM degradation complements exogenous degradation). Without per-cell MMP2 tracking, ECM degradation would have to be global or random rather than spatially restricted to mesenchymal cells at the invasive front.

1.4 Phenotype Parameters (Derived)

These are not independently stored — they are **computed** each time step from gene states + Boolean activity states + combination rules + knob values. Listed here because modules read them and PhysiCell needs them set.

Variable	Type	Tumor	Stromal	Derived from	Description & Justification
proliferation_rate	Continuous (mapped from qualitative tier)	✓	✓	M4: gene states + checkpoints + ZEB1 penalty + hypoxia + pressure	The rate at which the cell attempts division. For tumor cells, this is typicall VERY HIGH (Knob 3a), penalized by ZEB1 (go-vs-grow) and mechanical pressure (contact inhibition). For CAFs moderate; for PSCs, near-zero. Maps to PhysiCell cycle.transition_rate. With this, cells don't divide and the tumor ne generates the mass, hypoxia, or paracri pressure that drives barrier formation.
apoptosis_rate	Continuous (mapped from qualitative tier)	✓	✓	M4: gene states + drug kill signal + ABCB1	The probability of death per time step. I tumor cells at baseline, very low (Knob 4). Increases when intracellular drug is high and ABCB1 is off. For stromal cel near-zero in v1.0. Maps to PhysiCell death.apoptosis_rate. Without drug-modulated apoptosis, the EA's interventions have no mechanism to kil tumor cells — every drug strategy woul produce zero benefit.
motility_speed	Continuous (mapped from EMT extent)	✓	✓	M5: ZEB1/CDH1 state + EMT extent	How fast the cell migrates per time step Epithelial tumor cells: low. Mesenchym tumor cells: high (scaled by Knob 5b extent tier). CAFs: mild motility. PSCs: near-zero. Maps to PhysiCell motility.migration_speed. Required f Anchor 6: the spatial restriction of EMT

Variable	Type	Tumor	Stromal	Derived from	Description & Justification
					only meaningful if mesenchymal cells actually *move* differently from epithelia cells.
adhesion_strength	Continuous (mapped from CDH1 state)	✓	✓	M5: CDH1 state	How strongly the cell adheres to neighbors. CDH1=ON: high adhesion (epithelial cells stick together, form cohesive mass). CDH1=OFF: low adhesion (mesenchymal cells detach, infiltrate stroma). Maps to PhysiCell mechanics.cell_cell_adhesion_strengt Required for the tumor to have architectural structure — a cohesive co with a loose invasive front, not a diffus cloud of independent cells.
secretion_rates	Struct {tgfb, shh}	✓	Partial	M2: gene states + HIF1A + cell type + activation mode	How much TGF-β and SHH the cell ad to the microenvironment per time step. Tumor cells: both, scaled by Knobs 1a/ amplified by HIF1A. CAFs: TGF-β onl (feedback). PSCs: zero. Maps to PhysiCell secretion.secretion_rate pe substrate. This is the variable that initia the entire barrier cascade — the first lin in the causal chain.
ecm_production_rates	Struct {ha, collagen}	—	✓ (CAF only)	M6: ACTA2 + GLI1 state	How much HA and collagen the CAF adds to the ECM density field per time step. PSCs: zero. CAFs: high. CAFs wi GLI1=ON: boosted. This is the variable

Variable	Type	Tumor	Stromal	Derived from	Description & Justification
					that physically builds the barrier. Witho it, activated CAFs would be inert — the activation switch (ACTA2) would flip b nothing would happen. Required for ev anchor that involves ECM density (1, 2 5, 7, 8, 9, 10).

1.5 Internal Clocks & Accumulators

Variable	Type	Tumor	Stromal	Module	Description & Justification
intracellular_drug	Continuous (0–1)	✓	—	M7	Accumulated drug inside the cell. Increases via uptake from the local drug field. Decreases via ABCB1-mediated efflux (when active) and natural decay. The kill signal sent to Module 4 is a function of this concentration, not the extracellular drug level. Required because drug efficacy depends on intracellular accumulation, not just delivery — a cell can be bathed in drug but survive if efflux keeps

Variable	Type	Tumor	Stromal	Module	Description & Justification
					intracellular levels below the kill threshold. This variable is what makes the ABCB1 mechanism meaningful.
time_since_drug_exposure	Continuous (hours)	✓	—	M7	Clock that starts counting when intracellular_drug first exceeds a stress threshold. Used to gate ABCB1 activation: efflux turns on only after this timer exceeds the delay mapped from Knob 7a. Required for the temporal dynamics of resistance induction — the "vulnerability window" the EA should exploit. Without this timer, resistance would be either instant or impossible, and the EA would never discover pulsed dosing strategies.
time_alive	Continuous (hours)	✓	✓	None (bookkeeping)	How long this cell has existed since creation (birth or initial seeding). Not read by any module in v1.0. Tracked for post-hoc analysis: correlating cell age

Variable	Type	Tumor	Stromal	Module	Description & Justification
					with survival, position, and phenotype in sanctuary populations (Anchor 10). Negligible storage cost; high diagnostic value.

Tier 2: Local Environment Variables (Per Voxel)

Tracked on the PhysiCell microenvironment grid. Each voxel (typically 20µm × 20µm in 2D) holds values for all fields. Updated by PDE diffusion/decay solvers and by cell-driven source/sink terms each time step.

2.1 Diffusible Substrates

Variable	Units (sim)	Boundary condition	Diffuses?	Decays?	Sources	Sinks	Descripti Justificat
oxygen	Normalized (0–1)	Dirichlet: fixed at boundary (normal tissue pO_2)	Yes	Yes (natural decay + cellular consumption)	Boundary voxels	All living cells (consumption proportional to metabolic rate)	The maste spatial variable. (concentra at a given location determine whether HIF1A activates (

Variable	Units (sim)	Boundary condition	Diffuses?	Decays?	Sources	Sinks	Descripti Justificat
							which cor EMT (M5 secretion amplificat (M2), and stress adaptatior (M7). The field is wh translates density in biological consequer denser EC → slower diffusion steeper gradient – more hypo → more barrier-building signals. Required Anchors 5 (central hypoxia), (spatial El and the

Variable	Units (sim)	Boundary condition	Diffuses?	Decays?	Sources	Sinks	Descripti Justificat
							hypoxia amplificat feedback (Loop 2). Without C a tracked there is nc spatial heterogen in cell bel — every c would bel identically regardless its positio relative to barrier.
tgfb	Normalized (0–1)	Dirichlet: 0 at boundary (sink)	Yes	Yes (natural decay)	Tumor cells (M2), activated CAFs (M2 via E12)	Boundary (absorption), cellular uptake (mild)	The prima stromagen signal. Diffuses f secreting outward through th domain. L concentra at a strom cell's posi

Variable	Units (sim)	Boundary condition	Diffuses?	Decays?	Sources	Sinks	Descripti Justificat
							determine activation probabilit (M3). Loc concentra at a tumor cell's posi drives EM via the SM independe arm (M5, and — in SMAD4-i cells — gr arrest (M4 E09). The TGF-β fie the communic channel between tu and strom Required Anchors 1 (self-assembly) (SMAD4 asymmetr (periphera

Variable	Units (sim)	Boundary condition	Diffuses?	Decays?	Sources	Sinks	Descripti Justificat
							EMT driv TGF-β gradient). Without T β as a diffusible the tumor- stroma paracrine doesn't ex and the ba cannot sel assemble.
shh	Normalized (0–1)	Dirichlet: 0 at boundary (sink)	Yes	Yes (natural decay)	Tumor cells only (M2)	Boundary (absorption)	The secon stromager signal. Diffuses f tumor cell into surroundi stroma. L concentra at a strom cell's posi activates ((M3), whi boosts EC productio

Variable	Units (sim)	Boundary condition	Diffuses?	Decays?	Sources	Sinks	Descripti Justificat
							(E11) and proliferati (E13). Un TGF-β, Sl has no dir effect on t cells (para only). Required specificall Anchor 3 (SHH paradox): EA or test scenario n be able to SHH secr to zero an observe th stroma is partially reduced (arm gone) not elimin (TGF-β a persists), that the tu accelerate Without S

Variable	Units (sim)	Boundary condition	Diffuses?	Decays?	Sources	Sinks	Descripti Justificati
							as a separ field from TGF-β, y can't mod two-arm stromal activation test SHH- specific interventi
drug	Normalized (0–1)	Dirichlet: set by EA/scenario at boundary (0 at baseline, >0 when drug is applied)	Yes	Yes (natural decay + cellular uptake)	Boundary voxels (when drug is applied)	Tumor cells (uptake → intracellular_drug), natural decay	The therapeuti agent. Ent from the boundary (simulatin vascular delivery), diffuses inward, a impeded b ECM den Local concentra at a tumor cell's posi drives upt and the ki

Variable	Units (sim)	Boundary condition	Diffuses?	Decays?	Sources	Sinks	Descripti Justificat
							cascade (M → M4). T drug field what mak barrier *ma* — withou drug tryin penetrate, ECM is ju inert mate Required Anchors 2 (drug diff drops), 7 (ECM degradatic drug), 9 (barrier predicts d failure), 1 (spatial sanctuary drug field serves as t EA's prim lever: the can't chan the tumor' genetics, b

Variable	Units (sim)	Boundary condition	Diffuses?	Decays?	Sources	Sinks	Descripti Justificati
							can chang when, wh and how r drug arriv

2.2 Structural Fields

Variable	Units (sim)	Boundary condition	Diffuses?	Decays?	Sources	Sinks
ecm_density	Normalized (0–1)	Dirichlet: 0 at boundary	**No** (ECM is immobile)	Slow natural turnover (very low decay rate)	Activated CAFs (M6: HAS2 + COL1A1 production), mechanical compaction (M8)	MMP2-active tumor cells (M5 → M6: local degradation), EA degradation interventions

Variable	Units (sim)	Boundary condition	Diffuses?	Decays?	Sources	Sinks

Variable	Units (sim)	Boundary condition	Diffuses?	Decays?	Sources	Sinks
ecm_ha_fraction	0–1 (fraction of ecm_density attributable to HA)	N/A (derived)	No	Tracks with ECM density	HAS2-active CAFs	Hyaluronidase interventions

Variable	Units (sim)	Boundary condition	Diffuses?	Decays?	Sources	Sinks

Variable	Units (sim)	Boundary condition	Diffuses?	Decays?	Sources	Sinks
ecm_collagen_fraction	0–1 (= 1 − ecm_ha_fraction)	N/A (derived)	No	Tracks with ECM density	COL1A1-active CAFs	Collagenase interventions

Variable	Units (sim)	Boundary condition	Diffuses?	Decays?	Sources	Sinks

2.3 Derived Local Properties

These are not independently tracked fields — they are **computed from** the fields above at each voxel, each time step. Listed because multiple modules read them.

Variable	Computed from	Description & Justification
local_diffusion_coefficient	ecm_density, ecm_ha_fraction	Effective diffusion rate for O_2, drug, TGF-β, and SHH at this voxel. Decreases as ECM density increases, with HA fraction weighting the diffusion-impeding component more heavily than collagen fraction. This is the mechanism by which the barrier blocks drug delivery (Anchor 2) and creates hypoxia (Anchor 5). Computed per-substrate if needed (drug molecules are larger than O_2 and may be impeded more by HA gel), but v1.0 can start with a single shared modifier and refine later.
local_mechanical_stiffness	ecm_density, ecm_collagen_fraction	Resistance to deformation at this voxel. Increases with ECM density, weighted toward collagen fraction. Read by Module 8 to determine how much solid stress a dividing tumor cell generates when pushing against this voxel's ECM. Also affects cell migration speed through ECM — stiffer matrix is harder to move through even for MMP2-equipped cells. Required for Anchor 3 (mechanical confinement that restrains the tumor).

Tier 3: Host-Level Variables (Global Scalars)

These describe the state of the entire simulation domain. Updated by aggregating across all cells and voxels. Not spatially resolved.

3.1 Population Metrics

Variable	Type	Updated	Description & Justification
total_tumor_cells	Integer	Each time step	Count of all living tumor cell agents. The most basic measure of tumor burden. Required for Anchors 3 (SHH inhibition → tumor grows), 4 (SMAD4-intact → tumor grows slower), 9 (barrier predicts drug failure → more survivors). Also used for termination conditions: if tumor cell count reaches zero, the simulation can end early (complete eradication — expected to be rare).
total_psc_count	Integer	Each time step	Count of living quiescent stellate cells. Decreases as PSCs activate into CAFs (they don't die, they convert). Tracks the remaining "fuel" for barrier expansion. When this reaches zero, no new CAFs can be recruited — the barrier can only be maintained by existing CAFs, not expanded. Important for determining whether the stroma has reached its maximum possible extent.
total_caf_count	Integer	Each time step	Count of living activated fibroblasts. Increases monotonically in v1.0 (irreversible activation, low CAF death). Required for Anchor 7 (do CAFs persist after ECM degradation — they must, for barrier rebuilding) and for Anchor 1 (CAF count should increase from zero as the barrier self-assembles).

Variable	Type	Updated	Description & Justification
tumor_caf_ratio	Float	Each time step (derived)	total_tumor_cells / total_caf_count. A proxy for the tumor-to-stroma balance. In PDAC, this ratio is unusually low (more stroma than tumor by volume) — often 20–30% tumor, 70–80% stroma. Tracking this ratio allows comparison to patient histology. If the simulation produces a high ratio (mostly tumor, little stroma), the paracrine loop is too weak. If the ratio is extremely low (almost all stroma, tiny tumor), the tumor is growing too slowly or CAFs are overproliferating. Not directly required by any specific anchor, but invaluable for calibration sanity checks.

3.2 Barrier Metrics

Variable	Type	Updated	Description & Justification
mean_peritumoral_ecm	Float (0–1)	Each time step	Average ECM density in the voxel shell immediately surrounding the outermost tumor cells. The operational definition of "barrier density." Required for Anchors 2 (must increase over time), 9 (must correlate with drug failure), and as the primary barrier-maturity readout. Computed by identifying the tumor boundary (outermost tumor cell positions), selecting voxels within a fixed radius outward from this boundary, and averaging their ecm_density.

Variable	Type	Updated	Description & Justification
max_ecm_density	Float (0–1)	Each time step	Highest ECM density value in any voxel in the domain. Tracks whether the barrier has reached saturation (biologically implausible density) or is still growing. If this hits 1.0, the ECM model may be overproducing — check CAF production rates and mechanical compaction. Useful as a ceiling sanity check rather than a direct anchor metric.
barrier_thickness	Float (µm)	Each time step	Radial distance from the outermost tumor cell to the first voxel where ecm_density drops below a low threshold (e.g., 0.1). Measures how deep the barrier extends outward from the tumor. Required for Anchor 2 (drug must penetrate through this thickness to reach tumor). Also useful for Anchor 3: barrier thickness under SHH inhibition should be less than control, confirming that the intervention actually thinned the stroma.

3.3 Drug & Treatment Metrics

Variable	Type	Updated	Description & Justification
drug_at_tumor_centroid	Float (0–1)	Each time step (when drug is present)	Drug concentration at the geometric center of the tumor cell population. The headline metric for drug penetration. Required for Anchor 2 (must decline as barrier matures) and Anchor 7 (must increase when ECM is degraded). Computed by finding the mean position of all tumor cells and sampling the drug field at that location.
drug_penetration_half_depth	Float (µm)	Each time step (when	Distance from the domain boundary at which drug concentration drops to 50% of boundary value. A

Variable	Type	Updated	Description & Justification
		drug is present)	standardized measure of how far drug gets before the barrier attenuates it. Required for Anchors 2, 8, and 9. More robust than drug_at_tumor_centroid because it doesn't depend on tumor position within the domain — it measures the barrier's diffusion-blocking capability directly.
fraction_tumor_cells_drug_exposed	Float (0–1)	Each time step (when drug is present)	Fraction of tumor cells with intracellular_drug above a minimum effective threshold. Measures how many tumor cells the drug actually reached at a concentration sufficient to matter. Required for Anchor 9 (this fraction should decrease as barrier density increases) and Anchor 10 (post-treatment, the unexposed fraction should map to the spatial sanctuary).
mean_intracellular_drug	Float (0–1)	Each time step (when drug is present)	Average intracellular_drug across all living tumor cells. A population-level summary of drug exposure. Useful for comparing arms in Anchor 7 (degrade+drug should produce higher mean intracellular drug than drug alone) and for fitness function computation.

Variable	Type	Updated	Description & Justification
fraction_abcb1_active	Float (0–1)	Each time step	Fraction of tumor cells with abcb1_active = ON. Tracks resistance emergence across the population. Should start at 0, increase during drug exposure (with delay), and plateau. Required for validating Knob 7a (induction delay): if this fraction jumps to 1.0 instantly upon drug exposure, the delay mechanism is broken.

3.4 Spatial Structure Metrics

Variable	Type	Updated	Description & Justification
tumor_radius	Float (μm)	Each time step	Maximum distance from tumor centroid to any tumor cell. Measures spatial extent of the tumor. Required for Anchor 3 (SHH-blocked tumor should have larger radius than control). Also useful for tracking the expansion rate: if radius plateaus, mechanical confinement is working; if it grows linearly, the barrier isn't generating sufficient solid stress.
emt_fraction_core	Float (0–1)	Each time step	Fraction of tumor cells with ZEB1=ON in the inner 50% of the tumor radius. Required for Anchor 6 (should be low — core remains epithelial).
emt_fraction_periphery	Float (0–1)	Each time step	Same, in the outer 50% of the tumor radius. Required for Anchor 6 (should be high — periphery is mesenchymal). The ratio emt_fraction_periphery / emt_fraction_core is the primary spatial EMT metric. If this ratio is ≈ 1, EMT is spatially uniform and the model is wrong.

Variable	Type	Updated	Description & Justification
o2_at_centroid	Float (0–1)	Each time step	Oxygen concentration at the tumor centroid. Required for Anchor 5 (must decline as barrier matures). Paired with o2_at_periphery (mean O_2 in cells within 50μm of tumor edge) to verify the gradient direction: centroid O_2 < periphery O_2 at mature barrier.
survivor_mean_ecm	Float (0–1)	Post-treatment	Mean ECM density at the positions of surviving tumor cells after drug exposure. Compared to mean ECM at positions of dead cells. Required for Anchor 10 (survivors should be in denser ECM). Only computed during post-treatment analysis, not every time step.

3.5 Simulation Control

Variable	Type	Updated	Description & Justification
sim_time	Float (hours)	Each time step	Current simulated time. Required for: gating drug delivery onset/offset, computing time_since_drug_exposure, defining early/mid/late measurement windows for Anchors 2 and 5, and enforcing the maximum simulation duration.
random_seed	Integer	Set once at initialization	Seed for all stochastic processes (cell placement, probabilistic death, motility direction noise). Required for replicate runs: same seed = same outcome (deterministic reproducibility). Different seeds across replicates allow testing whether anchor pass/fail criteria hold beyond stochastic noise. At least 3 seeds per condition (see Anchor Metrics doc).

Variable	Type	Updated	Description & Justification
intervention_state	Struct	Modified by EA or scenario	Tracks what interventions are currently active: drug boundary concentration, SHH inhibition status, ECM degradation events, efflux inhibitor status. The EA writes to this; modules read from it. Required because the EA's genome is expressed as a time-varying intervention schedule, not a static parameter set — the EA might pulse drug on/off, delay SHH inhibition, or stagger interventions. Without explicit intervention state tracking, the simulation can't execute complex temporal strategies.

Cross-Reference: Variables by Reality Anchor

Which variables must be correct for each anchor to be testable:

Anchor	Cell-level variables required	Environment variables required	Host-level variables required
1 (Self-assembly)	acta2_active, secretion_rates, ecm_production_rates	tgfb, shh, ecm_density	total_caf_count, mean_peritumoral_ecm
2 (Drug diffusion)	—	drug, ecm_density, local_diffusion_coefficient	drug_at_tumor_centroid, drug_penetration_half_depth, mean_peritumoral_ecm
3 (SHH paradox)	gli1_active	shh, ecm_density	total_tumor_cells, tumor_radius, mean_peritumoral_ecm
4 (SMAD4 asymmetry)	gene_state[SMAD4], proliferation_rate	tgfb	total_tumor_cells
5 (Central hypoxia)	hif1a_active	oxygen, ecm_density	o2_at_centroid
6 (Peripheral EMT)	zeb1_active, cdh1_expressed, mmp2_active	tgfb, oxygen	emt_fraction_core, emt_fraction_periphery
7 (ECM degradation)	abcb1_active, intracellular_drug	drug, ecm_density	mean_peritumoral_ecm, drug_at_tumor_centroid, total_tumor_cells
8 (HA vs. collagen)	—	ecm_density, ecm_ha_fraction, ecm_collagen_fraction	drug_penetration_half_depth, tumor_radius
9 (Barrier predicts	intracellular_drug	drug, ecm_density	mean_peritumoral_ecm, total_tumor_cells,

Anchor	Cell-level variables required	Environment variables required	Host-level variables required
failure)			fraction_tumor_cells_drug_exposed
10 (Spatial sanctuary)	cell_id, position, abcb1_active, intracellular_drug	drug, ecm_density	survivor_mean_ecm, total_tumor_cells, fraction_abcb1_active

Variable Budget

Tier	Count	Storage concern
Cell-level (per agent)	21 gene states + 8 Booleans + 6 derived floats + 3 accumulators + 4 identity fields = **~42 values per cell**	At 7,000 peak cells: ~294,000 values. Negligible.
Environment (per voxel)	4 diffusible substrates + 1 structural field + 2 component fractions + 2 derived properties = **~9 values per voxel**	At 500×500 grid (2D, 20μm voxels on 10mm domain): ~2.25M values. Dominated by PDE solver cost, not storage.
Host-level (global)	**~18 scalar metrics**	Negligible.

The computational bottleneck is PDE solving for 4 diffusible substrates, not variable storage. ECM density is technically not a PDE (no diffusion), but its effect on the other 4 PDEs' diffusion coefficients means it's coupled into every PDE solve step.

42 values per cell. 9 values per voxel. 18 global scalars. Every one is traceable to a module that reads it, a Reality Anchor that tests it, or both. Nothing is tracked for decoration.